

WWII GERMAN JET AND ROCKET AIRCRAFT

ALTHOUGH THE GERMANS ACHIEVED the first jet flight, with a Heinkel He 178, a week before the outbreak of WWII, they experienced many difficulties and delays in developing practical military jets. Experience led to essential design advances such as replacing the standard tail wheel on piston-engined aircraft with a nose wheel, to stop the jet efflux hitting the runway at takeoff. But producing a sufficiently reliable jet engine that gave enough thrust proved time-consuming. Junkers eventually made their Jumo 004 an effective turbojet engine and supplied it to Heinkel, Messerschmitt (whose Me 262 emerged as the most prominent jet aircraft of the war), and Arado. Jets remained essentially experimental aircraft, difficult to fly and unreliable, but were not without impact late in the war.



LATE ENTRY

Introduced in 1944, the jet-propelled Me 262A-1 fighter hit top speeds over 100mph (160kph) faster than conventional Allied fighters.

Arado Ar 234 Blitz

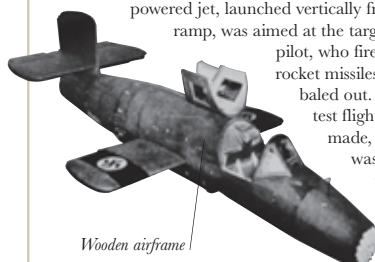
The world's first jet bomber (234A type shown) was originally designed as a long-range reconnaissance aircraft. When it went into service in August 1944, it finally provided the Germans with photographic intelligence which years of Allied air superiority had prevented. With the weight of the externally carried bomb slowing it down, the Arado carried out relatively few operations during the final defense of Germany.

ENGINE 2 x 1,980lb (900kg) thrust Junkers Jumo 004B turbojet	
WINGSPAN 46ft 3in (14.2m)	LENGTH 41ft 5in (12.7m)
TOP SPEED 461mph (742kph)	CREW 1
ARMAMENT 2 x 20mm MG151 cannon; 1 x 1,100lb (500kg) bomb	



Bachem Ba 349B-1 Natter

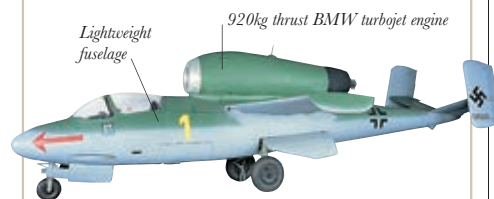
An emergency interceptor design developed in late 1944, the Bachem airframe was cheaply built of wood and disposable after one flight. The rocket-powered jet, launched vertically from a ramp, was aimed at the target by the pilot, who fired the rocket missiles and then baled out. Although test flights were made, the Natter was never used in action.



ENGINE 4,410lb (2,000kg) thrust Walter 509C liquid fuel rocket, plus 4 x 300kg (660lb) thrust Schmidding solid fuel rocket boosters	
WINGSPAN c.13ft 2in (c.4m)	LENGTH c.19ft 9in (c.6m)
TOP SPEED c.620mph (c.998kph)	CREW 1
ARMAMENT 24 x 73mm Hs 217 Föhn rocket missiles	

Heinkel He 162A-2

An emergency program for a lightweight mass-produced jet fighter was started in September 1944. Amazingly, the first 46 were delivered in February 1945, but by Germany's surrender, only a couple of dozen were operational. Effective in skilled hands – of which there was a shortage at that time – it recorded no "kills" in the few actions in which it participated.



ENGINE 2,028lb (920kg) thrust BMW 003E-2 turbojet	
WINGSPAN 23ft 7in (7.2m)	LENGTH 29ft 8in (9m)
TOP SPEED 522mph (840kph)	CREW 1
ARMAMENT 2 x 20mm Mauser MG151 cannon	

Heinkel He 178

The He 178 made the world's first jet-powered flight on August 27, 1939, nearly two years before the Whittle-engined Gloster E.28/39. Its designer, Dr. Hans Joachim Pabst von Ohain, began work on turbojets around the same time as Frank Whittle, but working for Heinkel (Whittle was a serving RAF officer) he was able to get a demonstration aircraft built sooner.

ENGINE 838lb (380kg) thrust Heinkel HeS 3b turbojet	
WINGSPAN 23ft 7in (7.2m)	LENGTH 24ft 6in (7.5m)
TOP SPEED c.373mph (c.600kph)	CREW 1
ARMAMENT None	



Messerschmitt Me 163 Komet

Development of this revolutionary rocket-powered interceptor started in 1938 under the title "Project X." On takeoff, the pilot jettisoned a small cart, and the aircraft climbed with phenomenal speed to engage Allied bombers with its heavy cannon armament.



ENGINE 3,750lb (1,701kg) thrust Walter liquid fuel rocket	
WINGSPAN 30ft 7in (9.3m)	LENGTH 18ft 8in (5.7m)
TOP SPEED 596mph (960kph)	CREW 1
ARMAMENT 2 x 30mm Rheinmetal-Borsig MK108 cannon	

Messerschmitt Me 262A-1a Schwalbe

While the excellent sweptwing Messerschmitt Me 262 – the first operational jet fighter in the world – was ready by 1941, it did not enter service until July 1944, due to continual engine problems. By then, they were heavily outnumbered by Allied aircraft.

ENGINE 2 x 1,980lb (900kg) thrust Junkers Jumo 004B turbojet	
WINGSPAN 40ft 11in (12.5m)	LENGTH 34ft 9in (10.6m)
TOP SPEED 540mph (870kph)	CREW 1
ARMAMENT 4 x 30mm Rheinmetal-Borsig MK108 cannon	

